

Basics of biofues

Any hydrocarbon fuel that is produced from an organic matter.

Produce less carbon dioxide, like bioethanol, dio diesel, biogas and biobutanol

Generations

1g : food crops

2g : non food crops

3g: specially made from algae

4g : GM algae and CCUS



National Policy on Biofuels

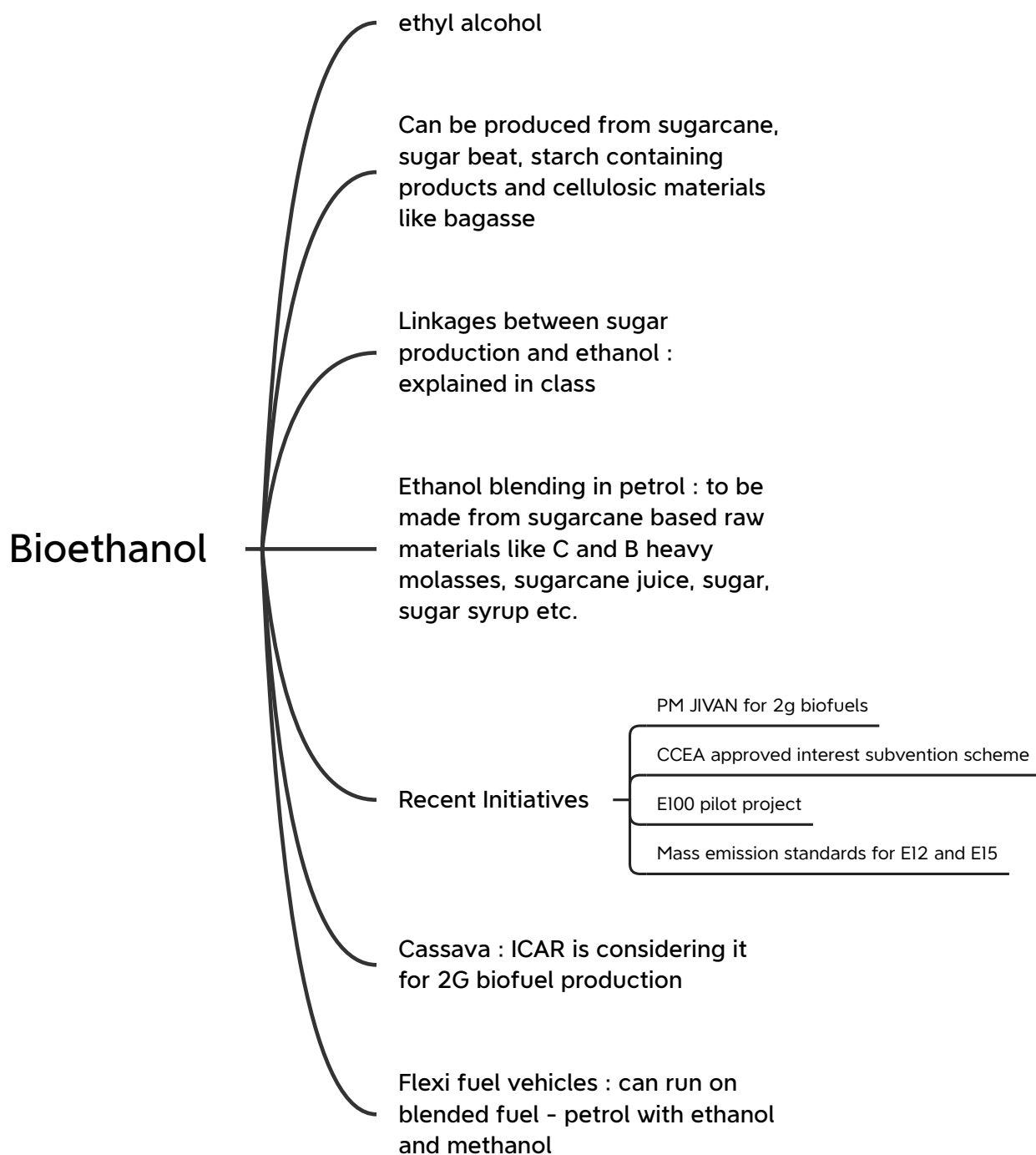
implemented by NBCC, chaired by minister of petroleum and natural gas.

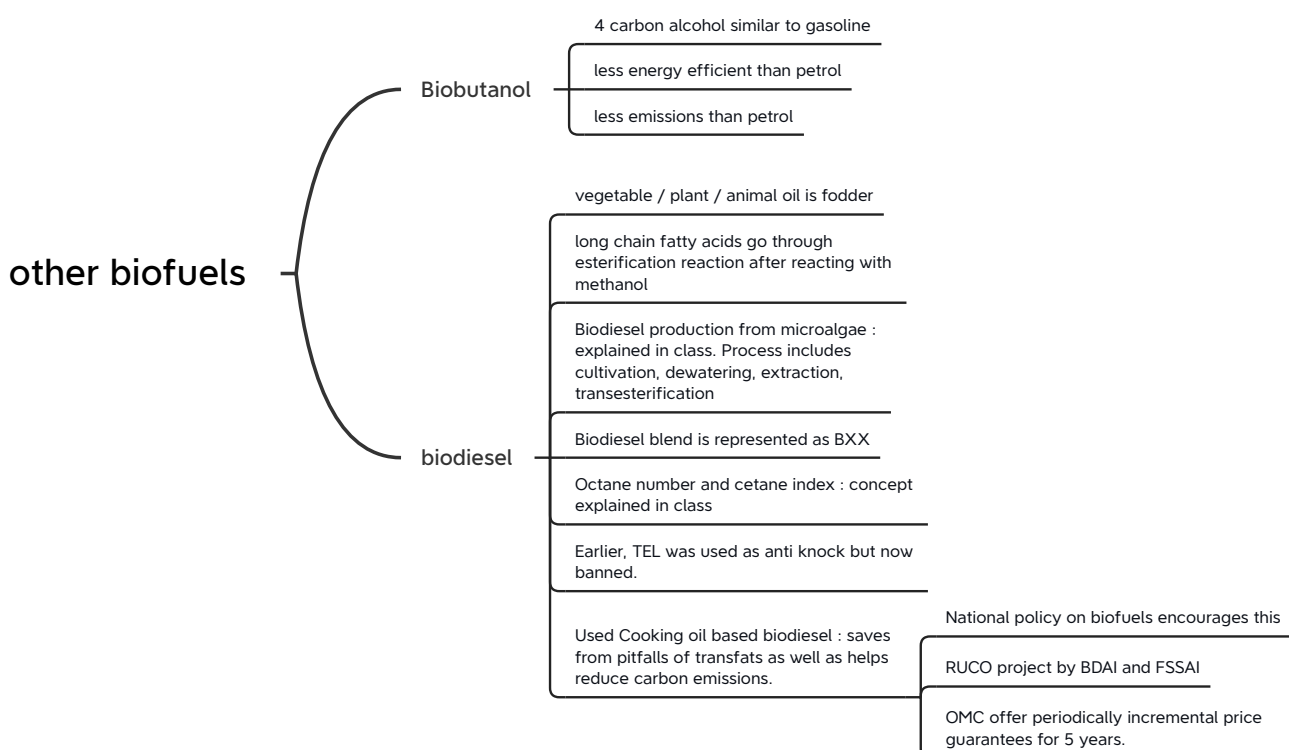
Target of E20 blending in petrol by 2025 and 5% biodiesel in diesel by 2030

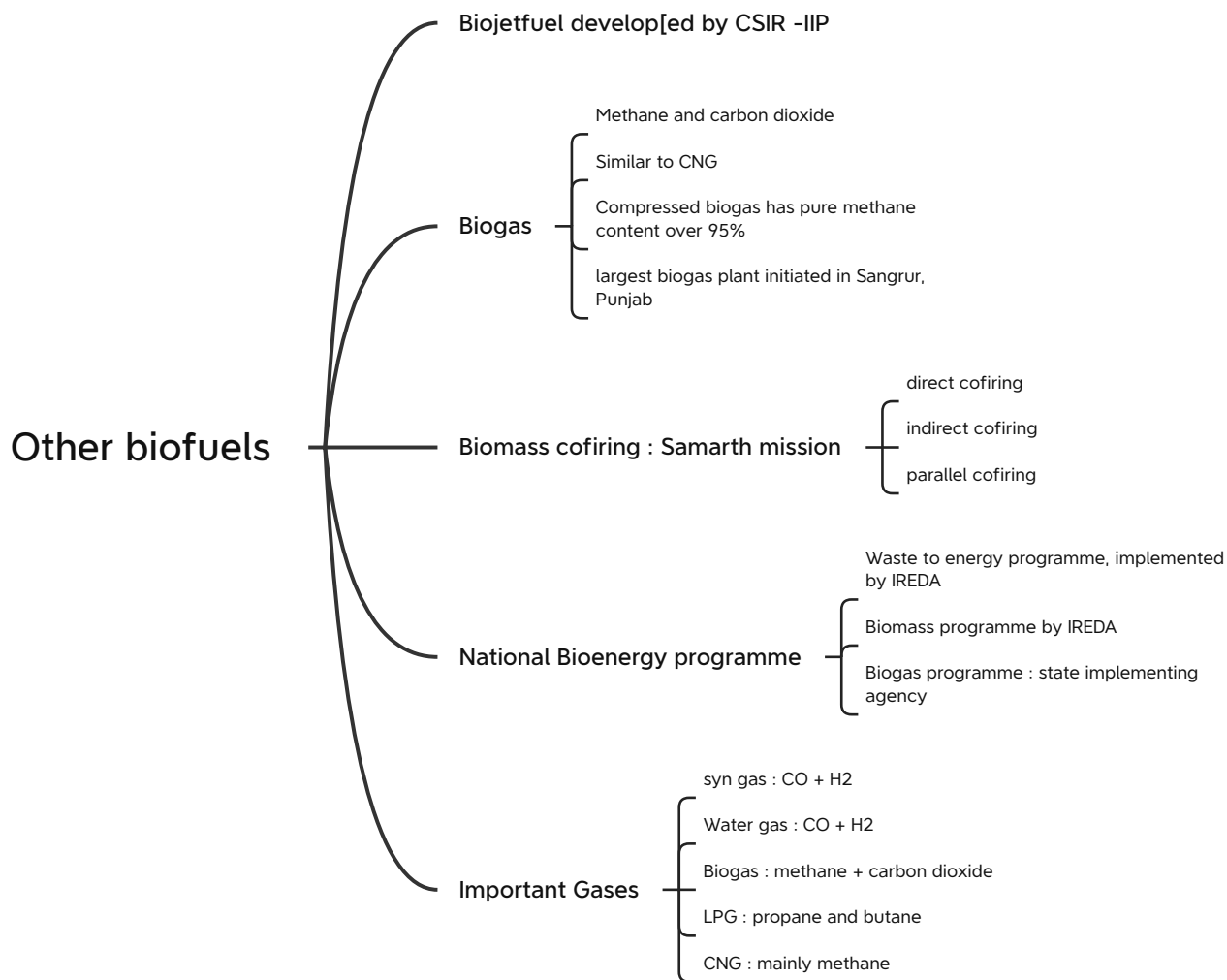
Classifies as basic, advanced and drop in fuels

Fodder for biofuels : sugarcane, damaged food grains, surplus food grains









Presented with xmind



Fuel Cell and other important initiatives

How fuel cell works : produces water and heat, explained in class

hydrogen enters at anode, positively charged particles cross the membrane to reach the cathode, electrons travel through another circuit to produce electricity. Water and heat are byproducts

Roshini : first saline water lantern which uses sea water, by NIOS



Wind energy

Concept is same : turning the turbine to generate AC power

locations : coastal, hilly, islands, deserts

Wind aero generator : small wind electric generator having capacity of 30kw

Wind solar hybrid system : MoNRE has issued a policy in 2018

India is currently 4th in the world, target is 140GW by 2030, 29% has been met

Important wind farms : Muppandal, Jaisalmer

Offshore wind energy policy by MoNRE

Global Offshore wind alliance : IRENA and GWEC



Small Hydropower

capacity of 25MW or below

earlier ministry of power, now
MNRE

Penstock : pipe through which
water flows and turns the turbine

Only local application, less power
generation and seasonal



Tidal Energy

using water difference in tides to move water across a turbine

requires high and low tide at regular intervals

Minimum height difference must be 5 meters

In india : gulf of kathch, gulf of khambhat and sundarbans delta (durgaduani creek project was stopped)



OTEC plant

temperature gradient between the surface and depth of the ocean is utilised to generate electricity and desalinate ocean water

difference must be atleast 20°C

open and closed cycle

India has relatively high potential of OTEC

NIOT (under MoES) to setup OTEC plant at kavaratti, lakshadweep with indigenous technology

Sindhuja : ocean wave energy converter by IIT madras



Geothermal energy

ONGC started drilling in Puga (ladakh) for geothermal energy

Uses hydrothermal resources.

India has high potential

India's geothermal provinces and Sites : map in slides



Deep Sea Mining

Remember UNCLOS limits

54% of the world's ocean is deep sea

permission must be taken from international seabed authority

Deep sea has critical minerals like Polymetallic nodules, massive sulphides and cobalt rich ferromanganese crusts

ISA

formed under UNCLOS

India is a member

effective protection of marine environment

UNCLOS has also established UNCLCS and ITLOS



Coal bed Methane

form of natural gas found in coal seams

extraction through use of water :
Explained in class

change in pressure alters the porosity of coal and methane is released, however there is a risk of leakage



Coal Gasification

process of converting coal gas into synthesis gas

done through partial oxidation

leads to reduced sulfur emissions

Ash and other byproducts are trapped at the bottom of the gasifier

Can be insitu or exsitu

clean coal technology, fresh water is recycled. Hydrogen, methanol and fertilisers can be formed through coal gasification

National Coal Gasification mission by Ministry of Coal

Interest rate subvention

Import duty exemption

Subtopic 3



